Don't Reward Lucky Guesses: Counterfactual World Pairs Teach Embodied Agents When to Look and When to Change Their Minds

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Abstract

Embodied vision-language agents should gather evidence when their observations cannot yet determine the right goal, and then change or keep that goal once the evidence arrives. Reinforcement learning makes this hard to teach: when credit is computed only in the world that happened to be realized, an agent that commits without looking is rewarded whenever its guess happens to be right. We construct pairs of simulated worlds that are consistent with exactly the same observation history but may call for different goals, and require behavioral divergence only after discriminating evidence has entered the agent's input. These pairs let us assign credit to evidence-seeking decisions over the worlds the agent cannot yet tell apart, which keeps the expected credit unchanged while removing the variance that comes from which world happened to be realized, and they let us supervise how the goal should be updated once the worlds can be told apart. Every pair is checked for deterministic restoration, physical validity, a unique target in each world, an executable success witness, and pixel-level information consistency. We compare against training on the same pairs without cross-world terms, same-world look-versus-commit branching, outcome-only reinforcement learning, and fixed and heuristic observation policies, and report success as a function of observation cost across different rates of world change.

1. Introduction

Consider an embodied agent asked to bring "the apple in the fridge." In most homes the apple is where it usually is, and an agent that walks to its habitual location will often succeed. Sometimes, however, someone has moved it, and the only way to know is to open the fridge and look. A competent agent should spend a little effort to check when its current observations cannot settle the question, and it should then act on what it sees: change its goal when the evidence says so, and keep it when the evidence does not.

Vision-language models (VLMs) are increasingly trained as such agents with reinforcement learning (RL) in simulation [10, 20, 21]. The usual learning signal, however, is computed in the single world that the episode happened to sample. If the agent commits without looking and the apple happens to be in its usual place, the episode succeeds and the premature commitment is reinforced; if the apple was moved, the same decision, taken on the same observations, is punished. The agent's inputs were identical in both cases, so the difference in credit carries no information the agent could have used. Averaged over many episodes the signal is still correct in expectation, but it is noisy exactly where it matters, and it never tells the agent what it should have done with the evidence it did not collect.

We propose to remove this noise at its source. Instead of sampling one world per episode, we construct *pairs* of simulated worlds that agree on everything the agent has observed so far but may differ in the hidden fact that determines the goal. While the agent cannot tell the two worlds apart, it acts once and receives credit averaged over both; once discriminating evidence enters its input, the two episodes are allowed, and required, to diverge. This gives two learning signals. First, *information-set credit* for evidence-seeking: the value of looking is compared with the value of committing now, averaged over the worlds consistent with the agent's information. Second, *post-evidence consistency* for the goal decision: after looking, the agent should switch targets in pairs whose goals differ (FLIP) and keep its target in pairs whose goals agree (STAY).

Building such pairs requires care. A pair is only useful if the two worlds really are indistinguishable to the agent up to the decision point, if each world has a unique and reachable target, and if the evidence becomes visible after looking. We therefore check every pair for deterministic restoration, physical validity, target uniqueness, an executable success witness, and pixel-level equality of the agent's inputs, and we discard pairs that fail any check.

We are careful about what the pairs do and do not buy. If worlds are sampled from the same prior, a same-world comparison between looking and committing, as in recent evidence-seeking navigation work [21], is also unbiased.

Our claim is narrower: at an equal rollout budget, pairing acts as stratified sampling over the hidden world, removing the between-world component of the variance of the credit, and it lets us balance FLIP and STAY cases and control how often the world changes. Our experiments are designed to test this claim directly.

Contributions.

- A formulation of evidence-seeking and goal updating in embodied VLM agents as credit assignment over information sets, with LOOK/FLIP/STAY cases defined relative to what the agent has observed (Sec. 3).
- A procedure for constructing and validating counterfactual world pairs in AI2-THOR [8], including pixel-level checks of information consistency and evidence visibility (Sec. 4).
- Information-set credit and post-evidence consistency objectives that plug into multi-turn PPO for VLM agents [20], with an analysis showing that they keep the expected credit unchanged while removing between-world variance (Sec. 3).
- A controlled comparison against outcome-only RL, the same paired data without cross-world terms, same-world look-versus-commit branching, and fixed and heuristic observation policies, at equal rollout budgets and across rates of world change (Sec. 5).

2. Related Work

RL for VLM agents and evidence seeking. VAGEN [20] trains VLM agents with multi-turn RL and world-model reasoning rewards, and mitigates templated reasoning with structured ground-truth evaluation and repetition penalties; we build on its trainer and include it as a baseline. GLANCE [10] rewards the discrepancy between predicted and observed frames as visual curiosity in the same environments. Closest to us, SeekVLN [21] observes that VLM navigation agents act confidently when key evidence is missing, and branches from the same state into evidence seeking and direct navigation to credit the seeking decision by the subsequent navigation return. Its credit is computed in the realized world; our pairs average credit over the worlds consistent with the agent's information and additionally supervise whether the goal should change after looking. Asking for help [16] and compare-before-commit navigation [26] address related decisions about when to act on uncertain information.

Counterfactual data and supervision. Counterfactual VLN [14] trains navigation agents to behave consistently across original and counterfactual environments. Grounding on Request [2] evaluates FLIP/STAY behavior under image–text conflicts and uses a counterfactual twin-margin objective on static data; our consistency terms are related but act only after evidence has been acquired through interaction. DEER-3D [23] edits 3D scenes to produce counterfactual question–answer pairs for grounding, and DUEL [15] perturbs textual statements in a self-play loop; both are static and do not involve acquiring evidence. Several works study how much static VLM reasoning depends on the image [5, 12]. For embodied policies, language interventions show that vision-language-action models often execute the original goal [6].

Environment design and robustness. Unsupervised environment design [3, 13], LLM-driven environment designers [7, 11], quality-diversity red-teaming of VLA policies [24, 25], and perturbation benchmarks [4] generate or search for environments that challenge an agent. We do not search for hard worlds; we construct pairs whose relation to the agent's information is known and verified, and use that relation in the learning signal.

Partial observability and counterfactual credit. Learning counterfactual world models under partial observability [27] studies when action sequences have distinguishable consequences. Our information-set credit is a stratified estimator over hidden world states, related to classical variance reduction [17] but applied to evidence-seeking decisions with verified counterfactual worlds.

3. Method

3.1. Setting

An episode takes place in a world w drawn from a set of simulated scenes. At step t the agent's input h_t consists of the instruction, all frames observed so far, and the textual description of the available actions. The instruction refers to a target object through a relation (*e.g.* "the mug next to the kettle"), resolved in the current state of w, which yields a unique correct target $g(w)$. The agent can take *observation actions* (move to a neighboring viewpoint, turn, open a container), each costing λ, or end the episode with $\textsc{select}(c)$ for a marked candidate c, which yields reward $\mathbb{1}[c = g(w)]$.

World pairs and cases. A world pair (w_0, w_1) shares the scene and differs by a single edit to one object, made with a complete pose list so that no other object moves. Let τ be the first step at which the agent's inputs in the two worlds differ, *i.e.* $h_t(w_0) = h_t(w_1)$ pixel for pixel for all $t < \tau$. Two binary properties define four cases: whether the evidence is already in the input at the start ($\tau = 0$, *direct*) or must be acquired by observation ($\tau > 0$, LOOK), and whether the correct target changes ($g(w_0) \neq g(w_1)$, FLIP) or not (STAY). LOOK→STAY pairs are as frequent

as LOOK→FLIP pairs, so that "looking means switching" is not a valid shortcut.

The constraint. Our objectives obey one rule: *whenever the agent's inputs in the two worlds are identical, the two worlds receive identical credit for the action taken.* Behavioral divergence is required only after discriminating evidence has entered the input. We enforce this with *lockstep* rollouts: for $t < \tau$ the policy samples one action from h_t and executes it in both worlds; from τ on, the two episodes continue independently.

3.2. Information-set credit

Let ρ be the prior probability of the edited world, $p(w_1) = \rho$ and $p(w_0) = 1 - \rho$. Let $G_t(w)$ be the return from step t in world w, including the cost of later observation actions and the final reward, and let

$$Q_c(w,t) = \sum_c \pi_\theta(\text{SELECT}(c) \mid h_t)\mathbb{1}[c = g(w)] \quad (1)$$

be the expected reward of committing immediately in world w, computed by scoring every candidate action without any additional rollout. For a step $t < \tau$, a same-world comparison of looking against committing [21] uses only the realized world w^*,

$$A_t^{\text{same}} = G_t(w^*) - Q_c(w^*, t), \quad (2)$$

which rewards a premature SELECT whenever the guess happens to match $g(w^*)$. Information-set credit averages over the worlds the agent cannot yet distinguish:

$$A_t^{\text{IS}} = \sum_{w \in \{w_0, w_1\}} p(w)\big[G_t(w) - Q_c(w,t)\big]. \quad (3)$$

For a SELECT action taken at $t < \tau$, $G_t(w)$ is the immediate reward in each world. After τ, and for direct pairs, we use standard multi-turn GAE [17, 20]. By construction, A_t^{IS} is identical in both worlds for every $t < \tau$, so the constraint above holds.

What pairing changes. Write $X_w = G_t(w) - Q_c(w,t)$ for a continuation sampled in world w. If worlds are sampled from the same prior, A_t^{same} and A_t^{IS} have the same expectation. At an equal budget of two continuations and $\rho = \frac{1}{2}$, the mean of two same-world estimates has variance $\frac{1}{2}\big(\mathbb{E}_w[\text{Var}\, X_w] + \text{Var}_w(\mathbb{E}\, X_w)\big)$, whereas the paired estimate has variance $\frac{1}{2}\,\mathbb{E}_w[\text{Var}\, X_w]$. Pairing is stratified sampling over the hidden world: it removes the between-world term $\text{Var}_w(\mathbb{E}\, X_w)$, which is large precisely when committing early pays off in one world and fails in the other. For $\rho \neq \frac{1}{2}$ one rollout per world is no longer a proportional allocation and this guarantee does not hold; we report the effect of ρ empirically (Sec. 5). We do not claim that same-world credit is biased.

3.3. Post-evidence consistency

Once $t \geq \tau$, the inputs differ and the correct behavior depends on the case. Let π_w^{sel} denote the policy's distribution over candidates at the step where it selects in world w, with candidates aligned across worlds by object identity. For FLIP pairs we add a twin-margin term in each world,

$$\mathcal{L}_{\text{FLIP}} = \sum_w \max\big(0,\ m - \log \pi_w^{\text{sel}}(g(w)) + \log \pi_w^{\text{sel}}(g(\bar{w}))\big), \quad (4)$$

where $\bar{w}$ is the other world of the pair; for STAY pairs we penalize disagreement between the two target distributions,

$$\mathcal{L}_{\text{STAY}} = \tfrac{1}{2}\Big[\text{KL}\big(\pi_{w_0}^{\text{sel}} \,\|\, \pi_{w_1}^{\text{sel}}\big) + \text{KL}\big(\pi_{w_1}^{\text{sel}} \,\|\, \pi_{w_0}^{\text{sel}}\big)\Big]. \quad (5)$$

Neither term constrains the route, which may legitimately differ between worlds. The full objective is the PPO loss with the advantages above plus $\beta(\mathcal{L}_{\text{FLIP}} + \mathcal{L}_{\text{STAY}})$.

3.4. Training

We train with the multi-turn PPO [18] of VAGEN [20] on a VLM policy. Each training step uses the same number of rollouts for every method: outcome-only PPO and same-world credit sample 64 single-world episodes; paired methods sample 32 pairs, *i.e.* 64 episodes. All methods share the reward, observation cost λ, optimizer, number of steps, and evaluation protocol; they differ only in how advantages are computed and in the consistency term.

4. Experimental Setup

Environment and task. We use AI2-THOR [8] kitchen and living-room scenes, split by scene into training and held-out test sets. Each scene is discretized into a navigation graph of viewpoints (TODO: exact count from T01) connected when they are close and mutually reachable; the agent can face eight headings. At every step the current frame is overlaid with numeric marks on visible candidates of the target category and on openable containers [22], and the text lists the reachable neighbors and the available actions. Episodes end with SELECT or after a fixed turn limit.

Constructing and validating world pairs. For each scene we sample a start viewpoint, an instruction, and a single-object edit, and set both worlds in the same simulator process with complete object-pose lists. A pair is kept only if it passes every check: (i) restoring a state twice reproduces the same frame; (ii) the edited object settles within a small tolerance of its commanded pose and no other object moves; (iii) the instruction resolves to a unique target in each world; (iv) a privileged planner reaches and selects the target in each world within a step budget, and its action sequence is stored as a witness; (v) for LOOK pairs, the

agent's inputs are pixel-identical in both worlds before observation, while for direct pairs they already differ inside the relevant objects' masks; (vi) after observation, the frames differ in at least a minimum number of pixels inside the relevant objects' instance masks, by a margin well above rendering noise. We report discard rates by reason. Training pairs come only from training scenes; the test pairs are generated from held-out scenes and frozen before any training.

Model and training. The policy is Qwen2.5-VL-3B-Instruct [1], trained with the multi-turn PPO of VAGEN [20] on top of verl [19] and vLLM [9]. All methods share one frozen configuration (reward, λ, turn limit, batch of 64 episodes per step, number of steps); TODO: number of steps and GPUs per run from T00b/T04. Unless stated otherwise the training change rate is $\rho = 0.5$.

Methods compared. *Outcome-only PPO* samples single worlds and uses standard GAE. *Paired, independent* uses exactly the same lockstep pairs as our method but computes GAE in each world separately; it isolates the cross-world terms. *Same-world credit* samples single worlds and uses A_t^{same} before the evidence step, an adaptation of fork-based evidence seeking [21] to our task. *Information-set credit* uses Eq. (3); *ours* adds post-evidence consistency. Training-free references keep the base model's target choices but fix when it may commit: never look, always explore up to the budget, explore for a fixed number of actions, or explore until the anchor and a target-category candidate are visible (the last uses ground-truth masks and is privileged).

Metrics and protocol. We report success rate (SR), observation cost (observation actions per episode), success as a function of observation cost, the rate of premature commitment (selecting before the evidence step in LOOK pairs), post-evidence update accuracy separately for LOOK→FLIP and LOOK→STAY, and accuracy on direct FLIP and STAY pairs. Each metric is evaluated on frozen test pairs at test change rates of 0, 0.25, and 0.5. Trained methods are run with three seeds; we report the mean, standard deviation, and 95% confidence interval across seeds, and call a difference significant only when the confidence interval of the Welch difference excludes zero.

5. Results

5.1. Do the tasks require selective looking?

TODO(T03).

5.2. Main comparison

TODO(T06).

5.3. What changes after the evidence arrives?

TODO(T06).

5.4. Ablations

TODO(T07).

6. Limitations and Conclusion

Limitations. All experiments use a single simulator, discrete viewpoints, single-object edits, and a 3B-parameter policy; we do not evaluate on real robots. World pairs require a simulator that can restore and edit states deterministically and expose instance masks, which limits where the method applies directly. The variance argument for pairing holds for a balanced prior with one rollout per world; for other priors the benefit must be established empirically.

Conclusion. TODO(after G2): two or three sentences, written only from verified results.

LLM Usage Statement

We used LLM-based assistants in this work. Claude (Anthropic) was used to plan experiments, write task specifications, review experiment code, logs and results, and draft and edit the text of all sections. Codex (OpenAI) was used to write and debug experiment code, run experiments on the compute cluster, and retrieve bibliographic metadata. All experimental numbers in this paper are generated programmatically from logged experiment outputs, and every reference was checked against bibliographic databases (arXiv, DBLP, Semantic Scholar, Crossref). The authors designed the study, verified all results and claims, and take full responsibility for the content of this paper.

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